

Prepared for YZi Labs (formerly Binance Labs)

DeData Protocol

A compute-to-data marketplace where privacy budget is the price of data.

Live on opBNB Testnet (chain 5611)

2 contracts · 38 Foundry tests

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Sharing data is irreversible, so it stays locked

Sharing is irreversible

Once raw data is copied, the owner loses control and cannot prove who leaked it.

Buyers cannot verify

Buyers must trust a seller's claims about results, with no on-chain deadline or recourse.

Privacy has no price

Data deals charge for files, not for how much privacy each query consumes.

DeData: buyers never receive raw data. They buy differentially private query results, priced per unit of privacy budget (ϵ).

Three mechanisms make private data sellable

Priced privacy budget

Each dataset has a fixed budget of ϵ units. The price rises as the budget is spent:

cost = price $\times$ ϵ $\times$ total $\div$ average remaining

Scarce privacy costs more, and the budget is a hard cap.

Canary leak attribution

Every result carries buyer-unique canary records, committed on-chain as a Merkle root. A leaked canary proves which request it came from.

**Upheld leak: buyer stake split
50% provider, 30% reporter,
20% treasury.**

Model provenance

A soulbound receipt (ERC-721 with ERC-5192) links a model hash to up to 50 completed requests and their datasets.

One receipt per model, non-transferable.

Privacy engine today: clipped mean plus Laplace noise, scale = range $\div$ ($n \times \epsilon \times$ unit). Results use ECDH P-256 with AES-256-GCM and are decrypted in the buyer's browser.

Five steps, with an exit at every stage

1

Request

A staked buyer pays for ϵ units and supplies a public key.

2

Compute

The provider runs a noised query off-chain and encrypts the result to the buyer.

3

Commit

Within 2 days the provider posts the result hash and canary root with a 20% bond.

4

Challenge

For 24 hours the buyer may dispute with a 10% bond. The arbiter has 3 days to rule.

5

Settle

Anyone can finalize. The provider is paid minus a 2.5% fee and the bond is returned.

Failure paths are built in. No commit in 2 days: the buyer is refunded. Arbiter silent: the dispute expires and both bonds are returned. All payouts are pull-based.

Everything here can be verified today

2

contracts deployed on
opBNB Testnet

38

Foundry tests,
including 2 fuzzed
accounting tests

918

lines of Solidity across
both contracts

5

tools: CLI, provider
node, keeper bot, leak
scanner, web app

DeDataProtocol 0x5c6dA7E2B4D37Bc88E828c6512c93C659e5f1F1d (block 209310482)

DeDataProvenance 0x3b9b87018E3f99841f4323dC2357BD21075dfbde (block 209311173)

Inspect both on testnet.opbnbscan.com. Testnet only: no mainnet deployment, audit, users or revenue yet.

Both sides risk money, so honesty pays

Protocol fee	2.5% of every completed request, paid to the treasury
Provider bond	20% of the request amount, returned on honest settlement
Dispute bond	10% of the request amount, forfeited by the losing side
Time windows	Commit 2 days · Challenge 1 day · Resolution 3 days
Buyer stake	14-day unstake cooldown and a 90-day leak watch period

Why BNB Chain: payments and stakes are in BNB, settlement runs on low-cost opBNB, and dataset hashes reference BNB Greenfield objects.

Who is trusted for what, stated plainly

Is the compute honest?

Optimistic today. The buyer can dispute within 24 hours and the arbiter rules. If the provider cheated, the buyer gets a full refund plus the provider's bond.

Why trust the buyer?

Trust is not needed in full. Buyers lock a stake first, and every result holds canaries unique to that buyer. A proven leak slashes the stake, and half goes to the provider.

Can the arbiter take funds?

No direct withdrawal. The arbiter only picks the winner of a dispute or leak report. If it stays silent, cases expire and bonds are returned. Replacing it needs a 2-day timelock.

Stated plainly: the arbiter is a single address today. A dishonest ruling could still misdirect bonds, so an independent multisig is the first planned fix.

Privacy has a hard cap; proof is next

How strong is the privacy?

Every query adds Laplace noise scaled to ϵ . Values are clipped to a fixed range, which bounds how much one record can move the answer.

The total budget is a hard cap on how many queries can run.

Standard technique. This implementation is not yet audited.

Why not a TEE today?

A trusted execution environment, or zero-knowledge proofs, is the planned path to verifiable compute.

The economic layer (pricing, bonds, provenance receipts) sits above the compute method, so it works with either approach.

Research item, not a shipped feature.

We name our limits before you find them

Limitations we state openly

- The arbiter is a single address and a trust anchor.
- The provider creates the canaries, so a false report through a second account is possible.
- Canaries are separate fields in the result, so a leaker could strip them.
- Dataset ownership on Greenfield is not verified on-chain.

Planned next steps

- Move the arbiter to an independent multisig.
- Commission an external security audit before any mainnet launch.
- Research verifiable compute, such as TEE or zero-knowledge proofs.
- Run a pilot with the first real data providers.

Three of these four limits are already documented in the repository, and the false-report case is covered by a test.

A working prototype, ready for a real pilot

Traction, stated honestly

No users or revenue yet. The protocol runs on testnet only.

What exists: two deployed contracts, 38 tests and working tooling that anyone can verify.

What we need

1. Introductions to real data providers
2. An audit partner
3. Feedback on the design

Funding would pay for the audit and a first pilot.

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